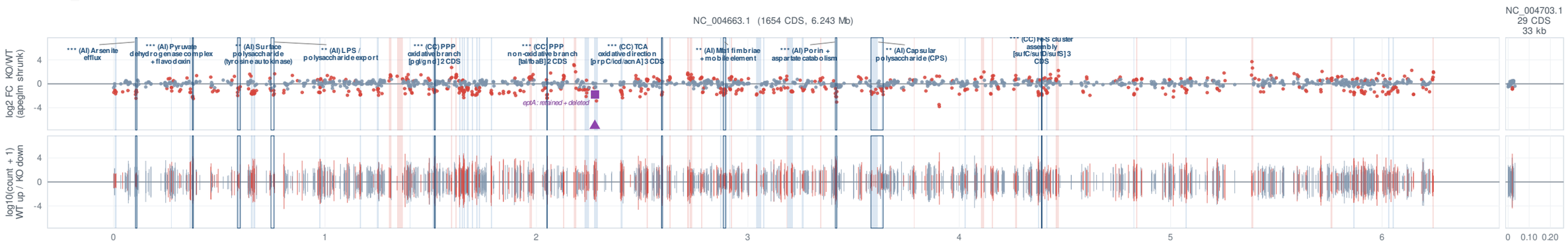
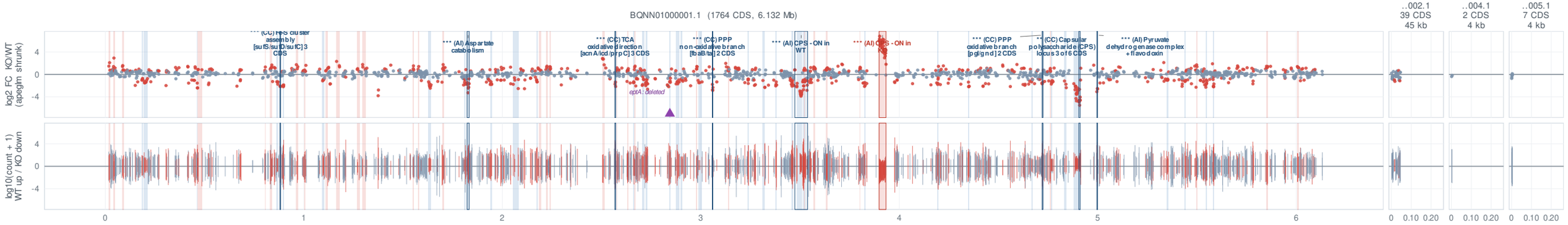
Regions called with AI and operons detected by CC - scheme (B) per strain

53 regions in total. "(AI)" = 17 regions called in the claude.ai discussion by reading the product names; "(CC)" = 36 operon-like blocks worked out with Claude Code from the DESeq2 tables of stages 4-6, with the counts giving matched genes out of the CDS inside the box. Where the two overlapped the CC coordinates were kept and the AI one dropped. HOW A (CC) BLOCK IS MADE: genes whose name or product matches the rule written for that function are chained together whenever the next one starts within 10 kb, and a chain is kept when it holds 2 or more of them (3 for ribosomal proteins) and at least one has padj < 0.05. Strand, intergenic distance and whatever genes happen to lie in between are NOT part of that rule, so a block is NOT evidence of a single transcription unit: of the 36 blocks drawn, 13 have unrelated genes inside them, 4 mix strands and 9 contain a gap of more than 200 bp (a real operon usually has gaps of tens of bp). Read a block as "genes of the same function sitting close together", nothing more. Box = the region, colour = direction (red = up in KO, blue = down in KO). The stars in front of each label say how sure the call is. *** = the product name itself states the function (AI rows), or the block is made of named members of a unit that was measured in stages 4-6 (CC rows). ** = the function is read indirectly from the group of products around it (AI rows), or the item was still listed as open at the end of stage 6 (CC rows: CPS). * = sequence similarity only, not yet verified. The two sets were scored separately, so the wording behind the stars is not identical between AI and CC rows. The purple square or triangle is drawn by the base figure and marks where eptA sits: a triangle means the log2 FC of that gene runs off the y axis, a square means it fits on it - the shape says nothing about retained vs deleted. The retained and deleted features are only 364-880 bp apart, under one pixel here, so they are named together in a single label. No fill is added, so the pale red/blue bands underneath are still the automatic ones (3+ consecutive DEGs in the same direction).

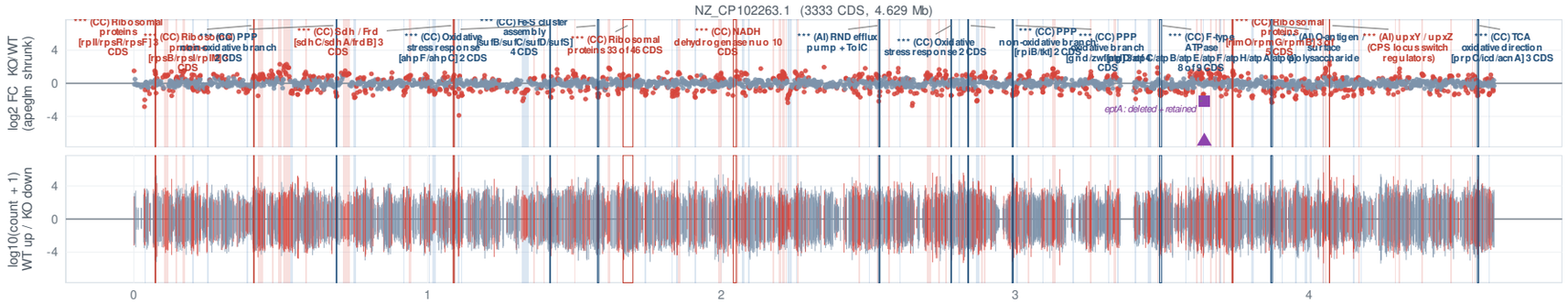
Bth_VPI



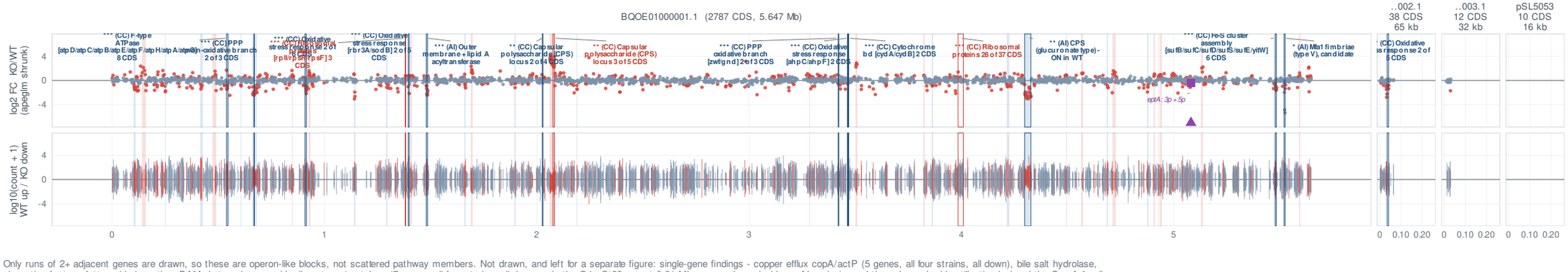
Bth_St8



Bun_8492



Odo_St23



Only runs of 2+ adjacent genes are drawn, so these are operon-like blocks, not scattered pathway members. Not drawn, and left for a separate figure: single-gene findings - copper efflux copA/actP (5 genes, all four strains, all down), bile salt hydrolase, elongation factors, fatty acid elongation, BAM, Lpt, and superoxide dismutase / catalase (7 genes, all four strains, all down; only the Odo_St23 one at 0.91 Mb appears here, inside an AI region) - and the polysaccharide utilization loci and the OmpA family.